MILITARY SPECIFICATION SHEET

METER, TIME TOTALIZING, MINIATURE, DIGITAL,
115 VAC, 50 TO 2400 HZ, SQUARE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirement for acquiring the meter described herein shall consist of this specification and the latest issue in effect of MIL-M-7793.

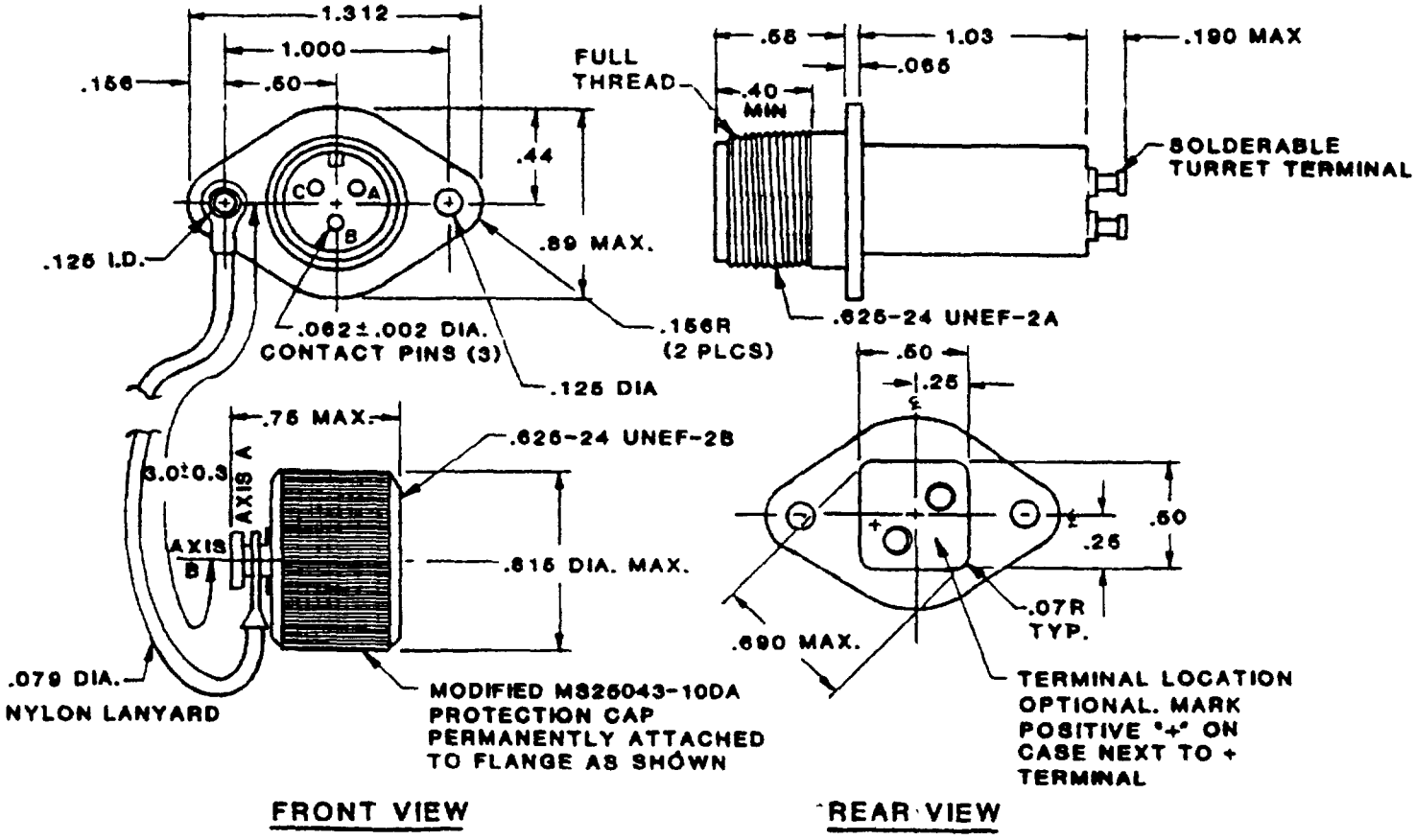


FIGURE 1.

AMSC N/A FSC 6645

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REQUIREMENTS:

1. Dimensions are in inches. Unless otherwise specified, tolerances are ±.02 for two-place decimals and ±.015 for three-place decimals.

2. The mounting flange, threaded extension and threaded protection cap shall be constructed of aluminum alloy with an olive drab chromate coating over cadmium plating.

3. The protection cap shall be in accordance with MS25043-12DA except for the finish and the means of attachment. The finish shall be in accordance with Requirement 2 and the metallic chain shall be replaced with a nylon lanyard.

4. Contacts shall meet all the requirements of silver plated solder contacts from MIL-C-5015.

Performance:

Dielectric strength. The meter shall be capable of withstanding 600 VRMS, 60 Hz (room) and 350 VRMS, 60 Hz (altitude).

Life accuracy: ±0.1% from -65° to +125°C and 75 to 150 VAC at 50 to 2400 Hz.

Power consumption: Power consumption at 115 VAC, 60 or 400 Hz, shall not exceed 50 milliwatts.

Operating temperature range: -65° to +125°C.

Part number: M7793/15-1.

Time range: Total readout of the meters shall be 99999.99 hours.

Meter output: The output from the meter shall be in a serial binary coded decimal format. The output impedance shall be 100K ± 10%. A logical "zero" shall be between 0.0 and +0.2 volts. A logical "one" shall be between +3.3 and +6.6 volts. The output pin assignments are shown in Figure 2 and the output data format is shown in Figure 3.

Transient protection: No temporary or permanent degradation shall be produced in the meter if the input voltage should increase to 180 VRMS at 50 to 2400 Hz for a period of 150 milliseconds maximum.

Durability of cap and threads: No mechanical or electrical degradation after 100 cycles in accordance with MIL-C-5015.

Voltage range: Operating voltage shall be from 75 to 150 VAC.

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Durability of connectors and contacts: No mechanical or electrical degradation after 100 cycles in accordance with MIL-C-5015. Operational accuracy shall be checked at the end of the test.

Weight: Weight shall be not greater than 1.0 ounce with the attached protection cap.

Tensile strength of lanyard: The lanyard shall withstand a tensile load of 25 pounds when applied to Axis "A" and Axis "B" as shown on Figure 1.

Qualification tests: Qualification tests will be conducted in accordance with MIL-M-7793, except that the dielectric strength, voltage, temperature and accuracy shall be as specified herein. Salt spray, moisture resistance, vibration, shock, electromagnetic compatibility and low temperature storage shall be conducted with the protective cap in place. Hermetic seal test is not applicable.

NOTES:

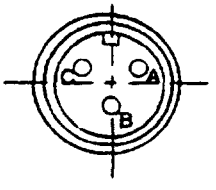
1. Elapsed time is read from the M7793/15-1 meter by mating the M7793/15-1 meter's connector with the M7793/12-1 reader's connector and operating the reader.

2. Digital readout and meter face plate per MIL-M-7793 are not applicable since the digital readout is provided by the separate M7793/12-1 reader.

3. The meter is designed to have an electrical ground and a continuous EMI/RFI shield with the threaded protection cap in place.

4. AC operation may be from either sinusoidal or square wave sources.

FRONT VIEW WITH PROTECTION CAP REMOVED.



A=INPUT FROM READER
B=COMMON FROM/TO READER
C=OUTPUT TO READER
PIN ARRANGEMENT PER MIL-C-5015
10SL-3

FIGURE 2. Readout pin assignment.

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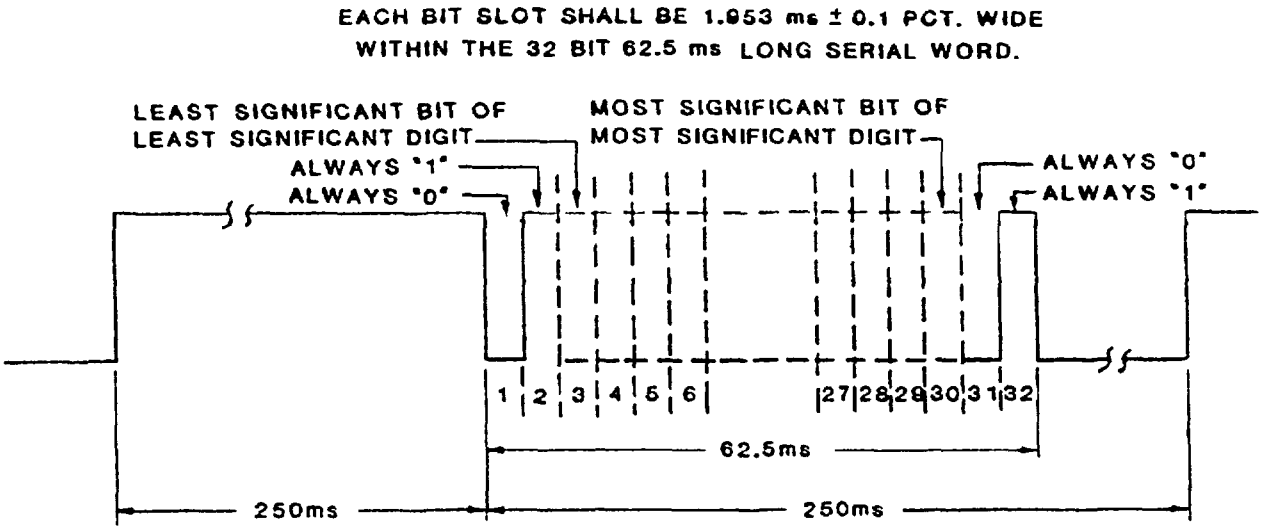


FIGURE 3. Output data format.

Custodians:
Navy - AS
Army -
Air Force - 99

Users:
Navy - SH, OS

Reviewer:
Army - MI

Preparing activity:
Navy - AS
(Project Number 6645-402)

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